

Diagnosis and treatment of plasmablastic lymphoma, plasmablastic myeloma and plasmablastic neoplasm: protocol for a scoping review

Hiroataka Mori¹, Masahiko Fukatsu¹, Hiroshi Ohkawara¹, Yuka Oka², Yuki Kataoka^{3,4,5,6}, Shunsuke Taito^{5,7}, Takayuki Ikezoe¹

1. Department of Hematology, Fukushima Medical University, Fukushima, 960-1295, Japan

2. Department of Diagnostic Pathology, Fukushima Medical University, Fukushima, 960-1295, Japan

3. Hospital Care Research Unit, Hyogo Prefectural Amagasaki General Medical Center, Higashinaniwa-cho 2-17-77, Amagasaki 660-8550 JAPAN.

4. Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Yoshida Konoe-cho, Sakyo-ku, Kyoto 606-8501 JAPAN

5. Systematic Review Workshop Peer Support Group (SRWS-PSG), Japan

6. Department of Respiratory Medicine, Hyogo Prefectural Amagasaki General Medical Center, Higashinaniwa-cho 2-17-77, Amagasaki 660-8550 JAPAN youkiti@gmail.com

7. Division of Rehabilitation, Department of Clinical Practice and Support, Hiroshima University Hospital, Kasumi 1-2-3, Minami-ku, Hiroshima, 734-8551 Japan

Abstract

Introduction

The differential diagnosis of plasmablastic lymphoma (PBL) from plasmablastic myeloma (PBM) is sometime very difficult. In such cases, the diagnosis of plasmablastic neoplasm (PBN) is acceptable. But clinicians face the question whether chemotherapies for PBL, PBM or others are effective for PBN and the question increases the diagnosis dilemma. Although evidence on the differential diagnosis and treatment of those three entities is compiling, there is no scoping review of different diagnosis and treatment for PBL, PBM, and PBN to answer those questions. The aim of this study is to develop an a priori protocol to undertake a scoping review of the

available evidence on the reproducibility of differential diagnosis and the treatment of PBL, PBM, and PBN.

Methods and analysis

For the conducting the scoping review protocol, we will use an scoping review methodology established in the Joanna Briggs Institute manual. This methodology has a five-stage approach: (1) identify the research question; (2) identify relevant studies; (3) select studies; (4) chart the data and (5) collating, summarize and report the results. We also will use the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews to assess the results.

Ethics and dissemination

Since this is a review of the literature, ethics approval is not indicated. We will disseminate the findings from this study in publications in peer-reviewed journals.

Background

Plasmablastic lymphoma (PBL) is classified into a very aggressive lymphoma presenting immunoblasts or plasmablasts and having plasma cell phenotypes (Swerdlow SH, 2017). PBL is diagnosed at advanced stages, resulting in poor prognosis and a median overall survival 8 months (Han, 2017). Due to the rare disease, the standard chemotherapy has not been established.

The differential diagnosis of PBL from plasmablastic myeloma (PBM) is sometime very difficult. PBL with plasmacytic differentiation has morphological and immunophenotypic features overlapping with those of PBM, especially extramedullary PBM. The absence of serum monoclonal protein and lytic bone lesion, the high proliferation index or positive of In situ hybridization for EBV-encoded small RNA (EBER) make the diagnosis of PBM unlikely (Delecluse et al., 1997, Vega et al., 2005, Gravelle et al., 2018). But some cases presents both characteristics of PBL and PBM (Cencini et al., 2016, Dittus et al., 2018). Therefore those feature cannot make clear differential diagnosis of PBL from PBM and diagnosis is sometime arbitrary (Castillo et al., 2015). In such cases, descriptive diagnosis "plasmablastic neoplasm, consistent with PBL or anaplastic plasmacytoma" is possible to be accepted (Swerdlow SH, 2017).

Clinicians face the question whether chemotherapies used for PBL, PBM, or others will be effective for PBL and the question increases the diagnostic dilemma. Currently most reviews are narrative reviews, especially of only PBL (Castillo et al., 2015, Harmon and Smith, 2016, Lopez and Abrisqueta, 2018), which are not suited to answer those dilemma and question. Therefore, the aim of this study is to conduct a scoping review to examine the available evidence on the reproducibility of differential diagnosis and the treatment of PBL, PBM, and PBN.

Methods

In preparing this protocol manuscript, the PRISMA extension for scoping reviews (PRISMA-ScR) (Tricco et al., 2018) statement was used. This proposed scoping review will adopt the scoping review framework by The Joanna Briggs Institute (JBI) (Peters et al., 2015), which involves the following five stage approach: Stage 1: Identifying the research question, Stage 2; Identifying relevant studies, Stage 3; Study selection, Stage 4; Charting the data, Stage 5; Collating, summarizing and reporting the results.

Stage 1: Identifying the research question

We are interested in the reproducibility of differential diagnosis and the treatment of PBL,

PBM, and PBN and the research questions are as followed:

1. “What pathological characteristics of patients are diagnosed as PBL, PBM and PBN?”
2. “What regimens are used and what is the prognosis of PBL, PBM and PBN?”

Stage 2: Identifying relevant studies

We used the Population, Concept, and Context (PCC) framework by the JBI for scoping review (Peters et al., 2015) to define the inclusion criteria as follows:

Participants

All studies that include patients with PBL, PBM or PBN according to the WHO classification (Swerdlow SH, 2017) or author's definition will be included. No restrictions regarding age, ethnicity, any other characteristics will be made.

Concept

We will review the existing literature on diagnosis and treatments of PBL, PBM and PBN.

Context

We will not limit the context to cultural factors such as geographic location, specific racial or gender or particular settings such as acute care, primary health care or the community.

A systematic search will perform using a three-step literature search. The first step will include an initial limited search of the MEDLINE. Then, we will carry out analyses of the text contained in the titles and abstracts and the index terms of retrieved papers. In second step to identify as many relevant studies as possible, the following databases will be searched: MEDLINE, CENTRAL, Embase, ClinicalTrials.gov, WHO-ICTRP. The search strategies for those databases were drafted by YK and HM and further refined through discussion (table 1). Third, we will check the reference lists of the included studies to search other resources. We will include any types of study design until April 2020 . Conference abstracts and systematic reviews will be excluded.

Table 1. Search strategy

MEDLINE search strategy

# 1	Plasmablastic Lymphoma[MeSH Terms]
# 2	plasmablastic lymphoma[tiab]
# 3	plasmablastic lymphomas[tiab]
# 4	plasmablastic neoplasms[tiab]

# 5	plasmablastic neoplasm[tiab]
# 6	#1 OR #2 OR #3 OR #4 OR #5
# 7	plasmablastic[tiab]
# 8	plasmablast[tiab]
# 9	#7 OR #8
# 10	Lymphoma[MeSH Terms]
# 11	lymphoma[tiab]
# 12	lymphomas[tiab]
# 13	#10 OR #11 OR #12
# 14	#9 AND #13
# 15	Multiple Myeloma[MeSH Terms]
# 16	myeloma[tiab]
# 17	myelomas[tiab]
# 18	Plasmacytoma[MeSH Terms]
# 19	plasmacytoma[tiab]
# 20	plasmacytomas[tiab]
# 21	plasmocytoma[tiab]
# 22	plasmocytomas[tiab]
# 23	#15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22
# 24	#9 AND #23
# 25	anaplastic[tiab]
# 26	#18 OR #19 OR #20 OR #21 OR #22
# 27	#25 AND #26
# 28	#6 OR #14 OR #24 OR #27

```

((((("plasmablastic lymphoma"[MeSH Terms] OR "plasmablastic
lymphoma"[Title/Abstract]) OR "plasmablastic lymphomas"[Title/Abstract]) OR
"plasmablastic neoplasms"[Title/Abstract]) ) OR (("plasmablastic"[Title/Abstract] OR
"plasmablast"[Title/Abstract]) AND (("lymphoma"[MeSH Terms] OR
"lymphoma"[Title/Abstract]) OR "lymphomas"[Title/Abstract]))) OR
(("plasmablastic"[Title/Abstract] OR "plasmablast"[Title/Abstract]) AND ((((((("multiple
myeloma"[MeSH Terms] OR "myeloma"[Title/Abstract]) OR "myelomas"[Title/Abstract])
OR "plasmacytoma"[MeSH Terms]) OR "plasmacytoma"[Title/Abstract]) OR
"plasmacytomas"[Title/Abstract]) OR "plasmocytoma"[Title/Abstract]) OR
"plasmocytomas"[Title/Abstract])))) OR ("anaplastic"[Title/Abstract] AND
((((("plasmacytoma"[MeSH Terms] OR "plasmacytoma"[Title/Abstract]) OR

```

"plasmacytomas"[Title/Abstract]) OR "plasmocytoma"[Title/Abstract]) OR
"plasmocytomas"[Title/Abstract]))

CENTRAL search strategy

- # 1 MeSH descriptor: [Plasmablastic Lymphoma] explode all trees
- # 2 ("plasmablastic lymphoma*"):ti, ab, kw
- # 3 (plasmablastic neoplasm*):ti, ab, kw
- # 4 #1 OR #2 OR #3
- # 5 (plasmablast*):ti, ab, kw
- # 6 MeSH descriptor: [Lymphoma] explode all trees
- # 7 (lymphoma*):ti, ab, kw
- # 8 #6 OR #7
- # 9 #5 AND #8
- # 10 MeSH descriptor: [Multiple Myeloma] explode all trees
- # 11 (myeloma*):ti, ab, kw
- # 12 MeSH descriptor: [Plasmacytoma] explode all trees
- # 13 (plasm*cytoma*):ti, ab, kw
- # 14 #10 OR #11 OR #12 OR #13
- # 15 #5 AND #14
- # 16 (anaplastic):ti, ab, kw
- # 17 #12 OR #13
- # 18 #16 AND #17
- # 19 #4 OR #9 OR #15 OR #18

(((((([mh "plasmablastic lymphoma"] OR "plasmablastic lymphoma":ti,ab) OR
"plasmablastic lymphomas":ti,ab) OR "plasmablastic neoplasms":ti,ab)) OR
((plasmablastic:ti,ab OR plasmablast:ti,ab) AND (([mh lymphoma] OR lymphoma:ti,ab)
OR lymphomas:ti,ab))) OR ((plasmablastic:ti,ab OR plasmablast:ti,ab) AND (((([mh
"multiple myeloma"] OR myeloma:ti,ab) OR myelomas:ti,ab) OR [mh plasmacytoma]) OR
plasmacytoma:ti,ab) OR plasmacytomas:ti,ab) OR plasmocytoma:ti,ab) OR
plasmocytomas:ti,ab))) OR (anaplastic:ti,ab AND ((([mh plasmacytoma] OR
plasmacytoma:ti,ab) OR plasmacytomas:ti,ab) OR plasmocytoma:ti,ab) OR
plasmocytomas:ti,ab))

Embase search strategy

- # 1 "EMB.EXACT.EXPLODE(Plasmablastic Lymphoma)

# 2	ab(plasmablastic lymphoma) OR ti(plasmablastic lymphoma)
# 3	ab(plasmablastic lymphomas) OR ti(plasmablastic lymphomas)
# 4	ab(plasmablastic neoplasms) OR ti(plasmablastic neoplasms)
# 5	ab(plasmablastic neoplasm) OR ti(plasmablastic neoplasm)
# 6	#1 OR #2 OR #3 OR #4 OR #5
# 7	ab(plasmablastic) OR ti(plasmablastic)
# 8	ab(plasmablast) OR ti(plasmablast)
# 9	#7 OR #8
# 10	"EMB.EXACT.EXPLODE(Lymphoma)
# 11	ab(lymphoma) OR ti(lymphoma)
# 12	ab(lymphomas) OR ti(lymphomas)
# 13	#10 OR #11 OR #12
# 14	#9 AND #13
# 15	"EMB.EXACT.EXPLODE(Multiple Myeloma)
# 16	ab(myeloma) OR ti(myeloma)
# 17	ab(myelomas) OR ti(myelomas)
# 18	"EMB.EXACT.EXPLODE(Plasmacytoma)
# 19	ab(plasmacytoma) OR ti(plasmacytoma)
# 20	ab(plasmacytomas) OR ti(plasmacytomas)
# 21	ab(plasmocytoma) OR ti(plasmocytoma)
# 22	ab(plasmocytomas) OR ti(plasmocytomas)
# 23	#19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26
# 24	#9 AND #23
# 25	ab(anaplastic) OR ti(anaplastic)
# 26	#18 OR #19 OR #20 OR #21 OR #22 OR #23
# 27	#25 AND #36
# 28	#6 OR #14 OR #24 OR #27

ClinicalTrials.gov

Condition and disease: plasmablastic lymphoma

WHO-ICTRP

Key word; plasmablastic lymphoma

Stage 3: Study selection

In the study selection process, we will adopt the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) flow diagram ([Moher et al., 2009](#)). We will apply the inclusion and exclusion criteria described above for the selection of studies. The process will be carried out independently by two researchers (HM and HO). When there is any disagreement, we reach an agreement by discussion. Reasons for excluded studies will be reported. We will conduct the pilot testing using a random sample of 25 citations for title and abstracts screening. If level of agreement does not reach 75% (or greater), the eligibility criteria will be modified. Modifications will be reported in detail in the full scoping review report.

Stage 4: Charting the data

We will also use the data extraction summary form of patients' characteristics (table 2) such as age, gender, performance status, Lugano stage, Ann Arbor stage, International Prognostic Index (IPI) score, tumor localization, status of HIV infection or immunodeficiency. We will count and calculate those data from case series or case reports. We will report the percentage or data in cohort studies or others. We will group those data by PBL, PBM and PBN.

Table 2. Summary form of patients' characteristics

	PBL	PBM	PBN
Age			
median			
< 65 years			
$\geq$ 65 years			
Gender			
male			
female			
Performance statu			
0-1			
2-4			
unknown			
Ann Arbor Stage			
I - II			
III-IV			
unknown			
IPI score			
low			
intermediate			
high			
unknown			
Tumor location			
Immune status			
HIV			
transplantation			
normal			

For the evaluation of the reproducibility of differential diagnosis, we will map author

and publication year, the diagnosis name, the used diagnosis criteria and the following pathological characteristics which are significant to make differential diagnosis(Delecluse et al., 1997,Vega et al., 2005, Gravelle et al., 2018): histological types (immunoblastic, plasmablastic, plasmacytic), EBER, ki-67 proliferation index $\geq 80\%$, myeloma signs such as serum or urinary monoclonal protein identified by protein electrophoresis or immunofixation, osteolytic lesion on skeletal radiography, CT or PET/MRI, ratio of involved-to-uninvolved serum free light chain(rFLC) ≥ 100 (elevated rFLC) or renal insufficiency with elevated rFLC; creatinine clearance <40 mL/minute or serum creatinine $> 2\text{mg/dL}$. We will also report other pathological characteristics such as CD20, CD30, CD38, CD56, CD79a, CD138, BCL2, BCL6, Cyclin D1, cytoplasmic immunoglobulin, kappa or lambda light chain, MUM1, PAX5 (table 3). We will group those data by PBL, PBM and PBN.

Table 3. Table of diagnosis and characteristics

Author and year diagnosis criteria			Historological type	Significant pathological characteristics							Other pathological characteristics															
				myeloma signs																						
				immunoblast	plasmablast	plasmacytic	EBER	ki-67 ≅ 80%	monoclonal protein	osteolytic lesion	elevated rFLC	renal insufficiency	prior plasma cell neoplasms	CD20	CD30	CD38	CD56	CD79a	CD138	BCL2	BCL6	Cyclin D1	cytoplasmic	immunoglobulin	kappa or lambda	light chain

The diagnosis follow chart described in Fig.1 was posed by hematologists (HM, MF and TI) and a pathologist (YK) through discussion based on the above studies. Those characteristics were described as follow:

patients with the following features will be classified into PBL

- positive for EBER and negative for myeloma signs
- negative for EBER, the ki-67 over 80% and negative for myeloma signs

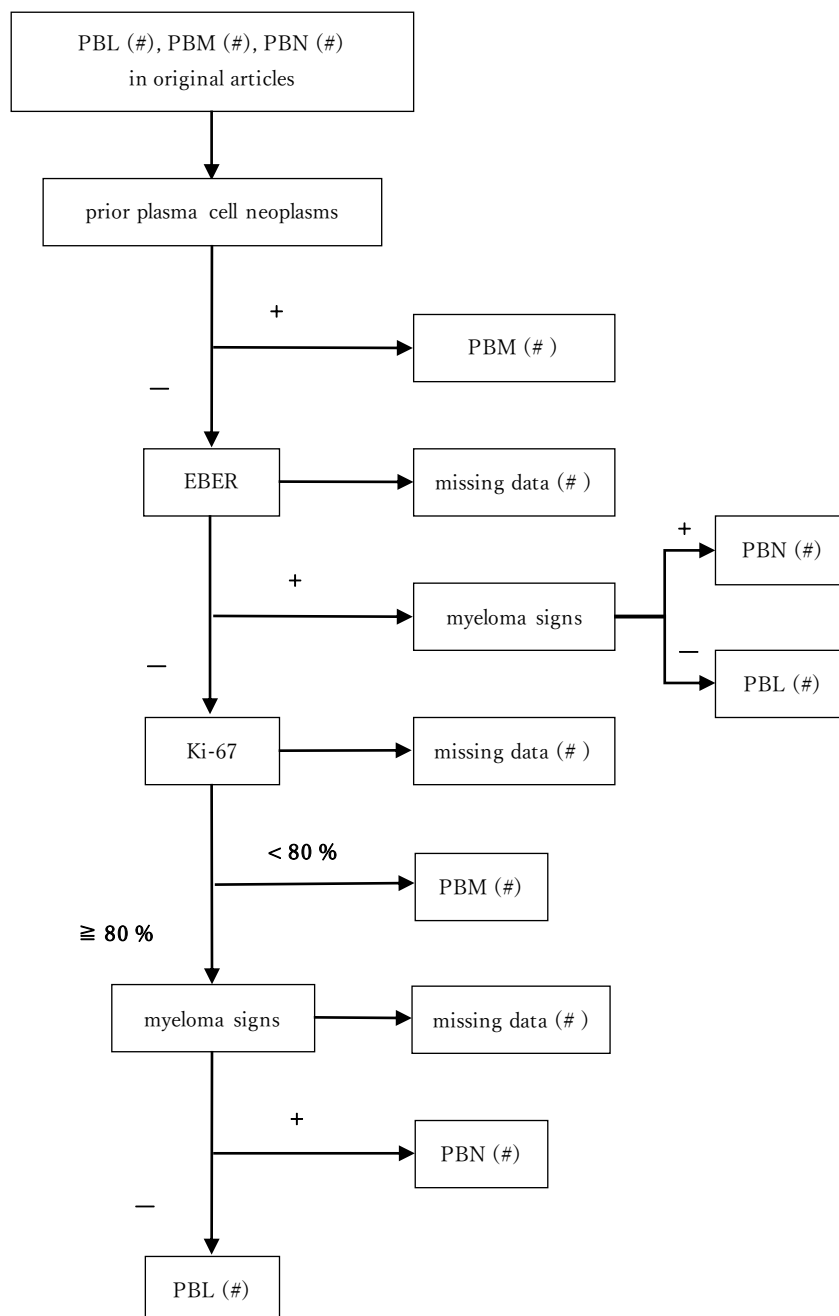
patients with the following features will be classified into PBM

- negative for EBER and ki-67 under 80%
- patients with prior plasma cell neoplasms

patients with the following features will be classified into PBN

- positive for EBER and myeloma signs
- negative for EBER, ki-67 over 80% and positive for myeloma signs

Figure 1. Diagnosis flow chart



#: numbers of patients or studies

+: positive, -: negative, <: less than, ≥: greater than or equal to

We will record the number of patients based on the flow chart. We will report the diagnosis criteria used in the observational studies. We will create a 3×3 table of the difference between the diagnosis in original articles and in flow chart. The number of patients will be described in the cells (table 4). We will count the number of PBN patients reported in the included articles from 2017 when WHO classification posed the diagnosis of PBN.

Table 4. 3×3 table of the difference between the diagnosis in original articles and in flow chart.

Diagnosis		Original articles		
		PBL	PBM	PBN
Flow chart	PBL			
	PBM			
	PBN			

The calibration of diagnosis using the flow chart will be carried out using a random sample of 25 citations for title and abstracts screening. If level of agreement does not reach 90%, the diagnosis flow chart will be modified. Modifications will be reported in detail in the full scoping review report.

For the prognosis, we are interested in the used regimens and evaluation of the efficacy and safety of regimens for each disease diagnosed based on the flow chart: therapy setting, intervention regimens, duration of the intervention, outcomes and adverse events. We will describe used regimens and classify them into the following intervention categories:

- Chemotherapies for lymphoma such as bendamustine (B), cyclophosphamide, doxorubicin, vincristine and prednisone (CHOP), etoposide, prednisone, vincristine, cyclophosphamide and doxorubicin (EPOCH), etoposide, cytarabine, cisplatin and methylprednisolone (ESHAP), ifosfamide, carboplatin and etoposide (ICE), pirarubicin, cyclophosphamide, vincristine and prednisolone (THP-COP), vincristine,

doxorubicin, cyclophosphamide, cytarabine, methotrexate and methotrexate (CODOX-M), cyclophosphamide, vincristine, doxorubicin and dexamethasone (Hyper-CVAD) etc.

- Chemotherapy for MM such as bortezomib (Bor), lenalidomide (L), thalidomide (Thal), bortezomib and dexamethasone (BD), lenalidomide and dexamethasone (LD), bortezomib, doxorubicin and dexamethasone (BAD), bortezomib, cyclophosphamide and dexamethasone (BCD), bortezomib, lenalidomide and dexamethasone (BLD), bortezomib, thalidomide, and dexamethasone (BTD) etc.
- Radiation therapy alone
- Surgical resection alone

When we described the combination chemotherapy with part of the chemotherapy for MM and part of chemotherapy for lymphoma, we used the hyphen for example, Bor-EPOCH. We will add other regimen names if they are identified in the process of charting the data.

We consider types of outcome as follow:

- Complete response (CR)
- Partial response (PR)
- Duration of response (DOR): the time between the overall response to therapy and subsequent disease progression, relapse or death.
- Grade 3 or 4 Adverse events (AEs)
- Relapse: progression disease or death

We will report the criteria used to evaluate outcomes and describe the details of response. We will report DOR as year and month or duration if studies report DOR in the style of year and month or duration. The number of courses conducted by a regimen will be described in the cells below regimen name. If relapse is progression disease, PD will be described. If death, D. Treatment and outcomes data will be grouped by PBL, PBM and PBN diagnosed based the diagnosis flow chart. The charting data form of treatment is shown in the table 5.

Table 5. Table of treatment and outcome

[illegible]

For each categorical or unknown data when reporting, a judgement will be made using one of three categories as follow:

- +: if the categorical data was described and positive
- -: if the categorical data was described and negative
- ?: if the report did not provide sufficient information

When we report data derived from case series, we will give each patient a number, for example patient 1, patient 2 and patient 3.

We will not contact the authors of included studies to request missing or additional data because we will also count the number of literatures which did not report necessary data. All results will be exported to Microsoft Excel 2016(Microsoft Corporation).

Stage 5: Collating, summarizing, and reporting the results

First, we will present the results of the search in the PRISMA flow chart. Second, we will organize the extracted data described above. The results of this scoping review will be presented as a map, in diagrammatic form, table and descriptive format.

Differences between the protocol and the overview

Changes in the methodology that need to be carried out throughout the study will be described in the results section.

Ethics and dissemination

This study will analyze only data of previously conducted studies, and will not involve any new studies performed by the authors. We will prepare the publication in accordance with PRISMA guideline and its adaptation for scoping reviews. We will publish our findings in peer-reviewed journals.

Patients and or public involvement

Patients and or public were not involved in the development of this protocol. The study group developed this study protocol without patient involvement.

Discussion

We believe that this scoping review will constitute foundational knowledge and provide evidence on the reproducibility of the differential diagnosis and treatment of PBL, PBM and PBN. Currently, most reviews revolving around differential diagnosis and treatments are narrative reviews ([Castillo et al., 2015](#), [Harmon and Smith, 2016](#), [Lopez and Abrisqueta, 2018](#)), which did not have the robust methodology. In addition, review research

methodologies should be used depending on the type as well as purpose of the research question ([Moher et al., 2015](#)). To our knowledge, the scoping review methodology is the best suited way to answer the research questions in this study. The results will provide insights into the uncertainty of evidence on the differential diagnosis and treatment of those three diseases, identify new questions and inform needed future researches.

Contribution

HM conceived the study. HM, YK and ST contributed to the protocol design and plan. HM and YK developed the literature search strategy. HM, MF, HO, YO, YK, ST and TI developed the charting data and drafted and revised manuscript.

Funding

No funding was obtained from any company.

Reference

- CASTILLO, J. J., BIBAS, M. & MIRANDA, R. N. 2015. The biology and treatment of plasmablastic lymphoma. *Blood*, 125, 2323-2330.
- CHEN, B.-J. & CHUANG, S.-S. 2020. Lymphoid Neoplasms With Plasmablastic Differentiation: A Comprehensive Review and Diagnostic Approaches. *Advances in anatomic pathology*, 27, 61-74.
- CENCINI, E., FABBRI, A., GUERRINI, S., MAZZEI, M. A., ROSSI, V. & BOCCHIA, M. 2016. Long-term remission in a case of plasmablastic lymphoma treated with COMP (cyclophosphamide, liposomal doxorubicin, vincristine, prednisone) and bortezomib. *European journal of haematology*, 96, 650-654.
- DELECLUSE, H. J., ANAGNOSTOPOULOS, I., DALLENBACH, F., HUMMEL, M., MARAFIOTI, T., SCHNEIDER, U., HUHN, D., SCHMIDT-WESTHAUSEN, A., REICHART, P. A., GROSS, U. & STEIN, H. 1997. Plasmablastic lymphomas of the oral cavity: a new entity associated with the human immunodeficiency virus infection. *Blood*, 89, 1413-1420.
- DITTUS, C., GROVER, N., ELLSWORTH, S., TAN, X. & PARK, S. I. 2018. Bortezomib in combination with dose-adjusted EPOCH (etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin) induces long-term survival in patients with plasmablastic lymphoma: a retrospective analysis. *Leukemia & lymphoma*, 59, 2121-2127.
- GRAVELLE, P., PÉRICART, S., TOSOLINI, M., FABIANI, B., COPPO, P., AMARA, N., TRAVERSE-GLÉHEN, A., VAN ACKER, N., BROUSSET, P., FOURNIE, J.-J. & LAURENT, C. 2018. EBV infection determines the immune hallmarks of plasmablastic lymphoma. *Oncoimmunology*, 7, e1486950-e1486950.
- HAN, X., DUAN, M., HU, L., ZHOU, D. & ZHANG, W. 2017. Plasmablastic lymphoma: Review of 60 Chinese cases and prognosis analysis. *Medicine (Baltimore)*, 96, e5981.
- HARMON, C. M. & SMITH, L. B. 2016. Plasmablastic Lymphoma: A Review of Clinicopathologic Features and Differential Diagnosis. *Arch Pathol Lab Med*, 140, 1074-8.
- LOPEZ, A. & ABRISQUETA, P. 2018. Plasmablastic lymphoma: current perspectives. *Blood Lymphat Cancer*, 8, 63-70.
- MOHER, D., STEWART, L. & SHEKELLE, P. 2015. All in the Family: systematic reviews, rapid reviews, scoping reviews, realist reviews, and more. *Syst Rev*, 4, 183.
- MOHER, D., LIBERATI, A., TETZLAFF, J. & ALTMAN, D. G. 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*, 6,

e1000097.

- PETERS, M. D. J., GODFREY, C. M., MCINERNEY, P., SOARES, C. B., KHALIL, H. & PARKER, D. 2015. *The Joanna Briggs Institute reviewers' manual 2015: methodology for JBI scoping reviews*.
- SWERDLOW SH, C. E., HARRIS NL, JAFFE ES, PILERI SA, STEIN H, THIELE J 2017. Plasmablastic lymphoma. *WHO Classification of Tumours of Haematopoietic and Lymphoid Tissues*, Revised 4th Edition, Volume 2.
- TRICCO, A. C., LILLIE, E., ZARIN, W., O'BRIEN, K. K., COLQUHOUN, H., LEVAC, D., MOHER, D., PETERS, M. D. J., HORSLEY, T., WEEKS, L., HEMPEL, S., AKL, E. A., CHANG, C., MCGOWAN, J., STEWART, L., HARTLING, L., ALDCROFT, A., WILSON, M. G., GARRITTY, C., LEWIN, S., GODFREY, C. M., MACDONALD, M. T., LANGLOIS, E. V., SOARES-WEISER, K., MORIARTY, J., CLIFFORD, T., TUNÇALP, Ö. & STRAUS, S. E. 2018. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann Intern Med*, 169, 467-473.
- VEGA, F., CHANG, C. C., MEDEIROS, L. J., UDDEN, M. M., CHO-VEGA, J. H., LAU, C. C., FINCH, C. J., VILCHEZ, R. A., MCGREGOR, D. & JORGENSEN, J. L. 2005. Plasmablastic lymphomas and plasmablastic plasma cell myelomas have nearly identical immunophenotypic profiles. *Mod Pathol*, 18, 806-15.